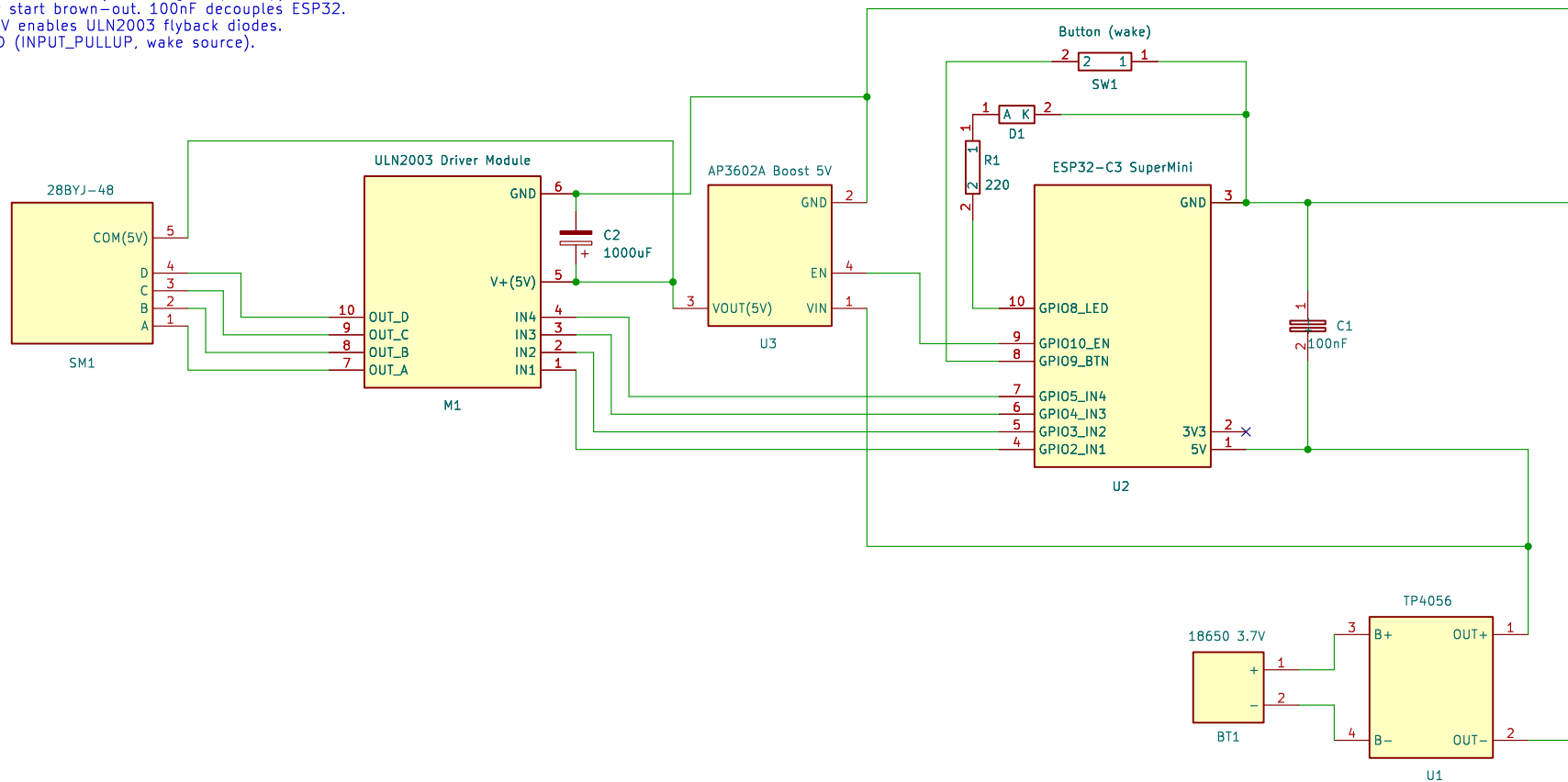


Cat Feeder power architecture:
 18650 -> TP4056 (charge/protect) -> OUT+ splits:
 (a) directly to ESP32-C3 5V/VIN pin (LDO makes 3.3V)
 (b) to AP3602A boost VIN -> VOUT 5V -> ULN2003 V+ and motor
 ESP32 GPIO10 controls boost EN (off during deep sleep).
 1000uF buffers motor start brown-out. 100nF decouples ESP32.
 COM of stepper -> 5V enables ULN2003 flyback diodes.
 Button GPIO9 -> GND (INPUT_PULLUP, wake source).



18650 -> TP4056 -> (ESP32-C3 5V) + (AP3602A boost -> 5V -> ULN2003 -> 28BYJ-48)

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File: cat_feeder.kicad_sch

Title: Cat Feeder - ESP32-C3 + 28BYJ-48

Size: A4 Date: 2026-06-16

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